

Mend-Amar Badral

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I'm a diligent and hard-working student with a passion for applying computational thinking to the most interesting problems. I aspire to contribute to the advancements in scientific discovery, machine learning and computer vision. I'm an avid Julia enthusiast and Python programmer.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.S. in Computer Science Engineering
Budapest, Hungary
Focus: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.S. in Applied Mathematics
Ulaanbaatar, Mongolia
Thesis: [Evaluating land degradation using image processing methods](#)

EXPERIENCE

Center of Mathematics Application Research Laboratory Oct, 2023 – June, 2025
Research Undergraduate Intern

- **Lab Leadership:** Selected as Undergraduate Lab Lead in final year; coordinated weekly research progress meetings and facilitated communication between students and faculty.
- **Infrastructure Management:** Managed and maintained the lab computer server infrastructure to ensure availability for team computational tasks.
- **Research:** Conducted research on image processing techniques to quantify plant types from drone imagery using segmentation methods (thresholding, K-means, Moran's I).
- **Documentation:** Authored a comprehensive laboratory handbook and technical documentation to standardize onboarding for future researchers.

PROJECTS

[Deep Learning Foundations and Concepts Book Figures Reproduction](#)

- Self-initiated project to reproduce key examples from C.Bishop and H.Bishop's *Deep Learning Foundations and Concepts* to gain theoretical and fundamental understanding of **polynomial regression** and **probability theory** based on early chapters of the book.
- Published reproducible Jupyter and Pluto.jl notebooks on GitHub, emphasizing clear explanation, interactivity, and visualization.

SKILLS

Programming Languages: Python, Julia, C, C++, R
ML Frameworks: PyTorch
Markup & Docs: Markdown, Quarto, L^AT_EX
Databases & Tools: SQLite, OpenCV, Git, Vim
Theoretical Background: Applied Mathematics, Digital Image Processing